DNS & DHCP TROUBLESHOOTING ASSIGNMENT

Hosts File • DHCP Gateway • DNS Diagnosis • APIPA • Common Issues

Student Name: _____ **Class:** _____
Date: _____ **Subject:** Networking

Safety note: The hosts-file exercise below temporarily changes name resolution only on your own computer. Use the reserved documentation IP address 192.0.2.123 as the deliberately incorrect test value. Restore the original hosts file immediately after testing. Do not change DNS for other users or devices.

1. Simulate a DNS resolution failure using the hosts file

The Windows hosts file can manually map a domain name to an IP address. A wrong entry can make the computer use an incorrect address instead of normal DNS resolution.

Steps:

1. Open Notepad as Administrator: search for Notepad, right-click it, and select Run as administrator.

2. In Notepad, choose File → Open and browse to:

```
C:\\Windows\\System32\\drivers\\etc\\hosts
```

3. Change the file-type selector from Text Documents to All Files. Add this temporary line at the bottom:

```
192.0.2.123    www.spotify.com
```

4. Save the file. Open Command Prompt and run:

```
ipconfig /flushdns
ping www.spotify.com
```

5. Record the output. It may show replies from 192.0.2.123 or timeouts / unreachable messages, depending on the test network. The important observation is that the name is mapped to the deliberately incorrect test IP. A ping timeout alone does not prove DNS failed.

Important: 192.0.2.0/24 is reserved for documentation and examples; it is not a real Spotify server address. After the test, reopen the hosts file as Administrator, remove only the test line you added, save it, and run **ipconfig /flushdns** again.

Actual ping output / error message:

_____

_____

Screenshot: Attach the Command Prompt result and, if requested, the temporary hosts entry (do not leave the wrong entry in place).

2. Configure DHCP with an incorrect default gateway in Packet Tracer

Example lab topology: one router, one switch, and one PC connected by Ethernet. Use a private lab subnet such as 192.168.10.0/24.

Example router DHCP configuration (the gateway is intentionally wrong for this exercise):

```
enable
configure terminal
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp pool LAB
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.254
 dns-server 8.8.8.8
end
show ip dhcp pool
show ip dhcp binding
```

Configure the router LAN interface with 192.168.10.1/24 and bring it up. On the PC, choose Desktop → IP Configuration → DHCP. The PC should receive an address from the pool and show 192.168.10.254 as its incorrect default gateway.

How to identify and fix:

1. On the PC, run **ipconfig /all** and check the Default Gateway field.
2. Compare it with the actual router LAN interface address.
3. On the router, inspect **show running-config** and the DHCP pool settings.
4. Correct the pool using **default-router 192.168.10.1**.
5. Renew the PC lease (or toggle DHCP off and on in Packet Tracer), then verify the gateway again. Test with **ping 192.168.10.1**.

My assigned IP / incorrect gateway / corrected gateway:

_____

_____

Screenshot: Attach the PC IP Configuration screen and router DHCP pool configuration.

3. Diagnose DNS when websites work by IP but not by name

Scenario: A client can reach a known server by IP address, but cannot open a website by its domain name. This often indicates a DNS configuration or DNS server problem.

Troubleshooting steps:

1. Check IP configuration and DNS server address:

```
ipconfig /all
```

2. Test basic network connectivity to the default gateway:

```
ping <default-gateway-IP>
```

3. Test connectivity to a known external IP address, if the lab network has internet access:

```
ping 8.8.8.8
```

4. Query DNS using nslookup:

```
nslookup www.example.com
nslookup www.example.com 8.8.8.8
```

5. If the DNS server is wrong or unreachable, correct the client DNS setting or the DHCP pool's DNS-server option. If using a local DNS server, check that its service is running and its records / forwarding settings are correct.

6. Clear the local DNS cache and test again:

```
ipconfig /flushdns
nslookup www.example.com
ping www.example.com
```

Possible solution: Configure a reachable, correct DNS server address (for example, the DNS server provided by the lab or ISP). Verify that DNS queries are allowed through the firewall. In a Packet Tracer lab, ensure the DNS service is enabled and the client has the correct DNS server IP.

Commands run and results:

_____

_____

_____

4. Troubleshoot a client receiving an APIPA address (169.254.x.x)

APIPA (Automatic Private IP Addressing) is a self-assigned IPv4 address in the 169.254.0.0/16 range. Windows may use it when it cannot obtain a valid address from a DHCP server. It usually allows only limited local-link communication, not normal routed network access.

Possible causes

• DHCP server is powered off, stopped, or unreachable.
• Ethernet cable, Wi-Fi connection, switch port, or VLAN is incorrect.
• DHCP scope has no free addresses (scope exhaustion).
• DHCP pool / scope is configured incorrectly.
• DHCP relay / IP helper configuration is missing or incorrect when the server is on another subnet.
• Firewall or network ACL blocks DHCP traffic (UDP ports 67 and 68).

Step-by-step resolution

1. Check the physical or wireless connection and link status.
2. Run **ipconfig /all**; confirm the client has a 169.254.x.x address and DHCP is enabled.
3. Try **ipconfig /release** followed by **ipconfig /renew**.
4. Ping the default gateway if a valid gateway is present; inspect switch port, VLAN, and cabling if not.
5. Check DHCP server status and confirm the correct scope is active.
6. Check available leases / scope capacity; expand the scope or remove stale leases if appropriate.
7. If the DHCP server is on another subnet, verify the router's DHCP relay / **ip helper-address** points to the correct server IP.
8. Check firewall / ACL rules for DHCP traffic, then renew the client lease and verify the new IP, subnet mask, gateway, and DNS values.

My troubleshooting notes:

_____

_____

_____

Useful commands:

```
ipconfig /all
ipconfig /release
```

```
ipconfig /renew
ping <default-gateway-IP>
nslookup www.example.com
```

5. Common DNS and DHCP issues table

The following examples are common issues researched for this assignment. Add your own lab observations in the final column if you performed the tests.

Common DNS issues

DNS issue	Symptoms	Possible cause	Recommended solution
DNS server unavailable	Names do not resolve; nslookup times out.	DNS service stopped, server unreachable, or firewall blocks DNS.	Check service status and IP connectivity; allow DNS on UDP/TCP
Incorrect DNS address	Websites fail by name but may work by IP.	Client or DHCP provides the wrong DNS server address.	Correct the DNS setting or DHCP DNS-server option; renew the
Wrong / missing DNS record	One domain resolves to the wrong IP or returns NXDOMAIN.	Incorrect A/CNAME record, typo, or stale zone data.	Correct the record, validate/reload the zone, and clear cache if ne

Common DHCP issues

DHCP issue	Symptoms	Possible cause	Recommended solution
DHCP scope exhausted	New clients get 169.254.x.x or no lease.	All addresses in the scope are in use.	Expand the scope safely, review leases, and remove stale reserv
DHCP server unavailable	Client cannot obtain a lease.	Server/service is down or network path is broken.	Start/check DHCP service; verify cabling, VLAN, routing, and ser
Incorrect DHCP options	Client gets an IP but cannot reach other networks or resolve names.	Wrong default gateway or DNS option in the scope.	Correct router/default-router and DNS options; renew the client le

Conclusion

In this assignment, I learned how to simulate a DNS mapping problem, identify an incorrect DHCP gateway, diagnose name-resolution failures, troubleshoot APIPA addresses, and compare common DNS and DHCP issues with their solutions.

Student Signature: _____ **Date:** _____